

AI BASED POSTURE ANALYSIS

Our Physio, Your Therapy



PROBLEM STATEMENT



UNAVAILABILITY OF PHYSIOTHERAPIST

According to WHO 2024 report, there are less than 1 physiotherapist per 10,000 people. Traditional physiotherapy depends on in-person observation which is time-consuming and inconsistent,



LACK OF COMMUNICATION

Summaries are either too technical for patients or too simplified for clinicians, leading to miscommunication and poor treatment adherence.



GENERATING CONTEXTUAL MEDICAL SUMMARIES

Build a custom model that generates perspective-aware medical summaries from Q&A pairs

SOLUTION

Posture Detection

Uses MediaPipe/OpenPose + deep learning to analyze exercise videos and physiotherapy movements in real time.

Posture Correction

Detects incorrect posture, joint misalignments, and risky movements, then suggests corrective actions instantly.



Summarization Model

Built from scratch (no API), optimized for medical text summarization.

Chatbot Assistant

Conversational interface that explains medical summaries, posture analysis results, and physiotherapy routines.

FEATURES



01

EXERCISES

Personalized physiotherapy exercise plans with guided instructions and real-time posture tracking to ensure correct form and safe recovery.



02

CONSULTATION

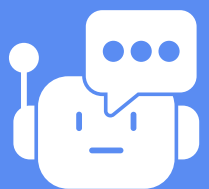
Virtual or in-person sessions with physiotherapists where patients can discuss their health, progress, and receive expert advice.



03

IMAGE ANALYSIS

AI-powered posture and movement analysis using computer vision to detect errors, measure improvements, and provide corrective feedback.



04

CHATBOT

An AI knowledge assistant that answers patient queries, explains conditions, tracks progress, and connects with physiotherapist data anytime.



TECHNICAL APPROACH

Data Preprocessing

Medical Q&A dataset cleaning, section classification, and text chunking.



Posture AI Integration

MediaPipe/ OpenPose pipeline, keypoint extraction, anomaly detection, recommendation engine.



Image to text

Optical Character Recognition(OCR)+ Natural Language Processing (NLP)



Embedding + Summarization Model:

Train transformer-based summarization model with LLM Model GPT4ALL fine-tuning for lightweight deployment.

Chatbot Integration

LangChain-based local RAG model + physiotherapy knowledge base + patient database.



WIREFRAME

AI-Powered Physiotherapy Chatbot

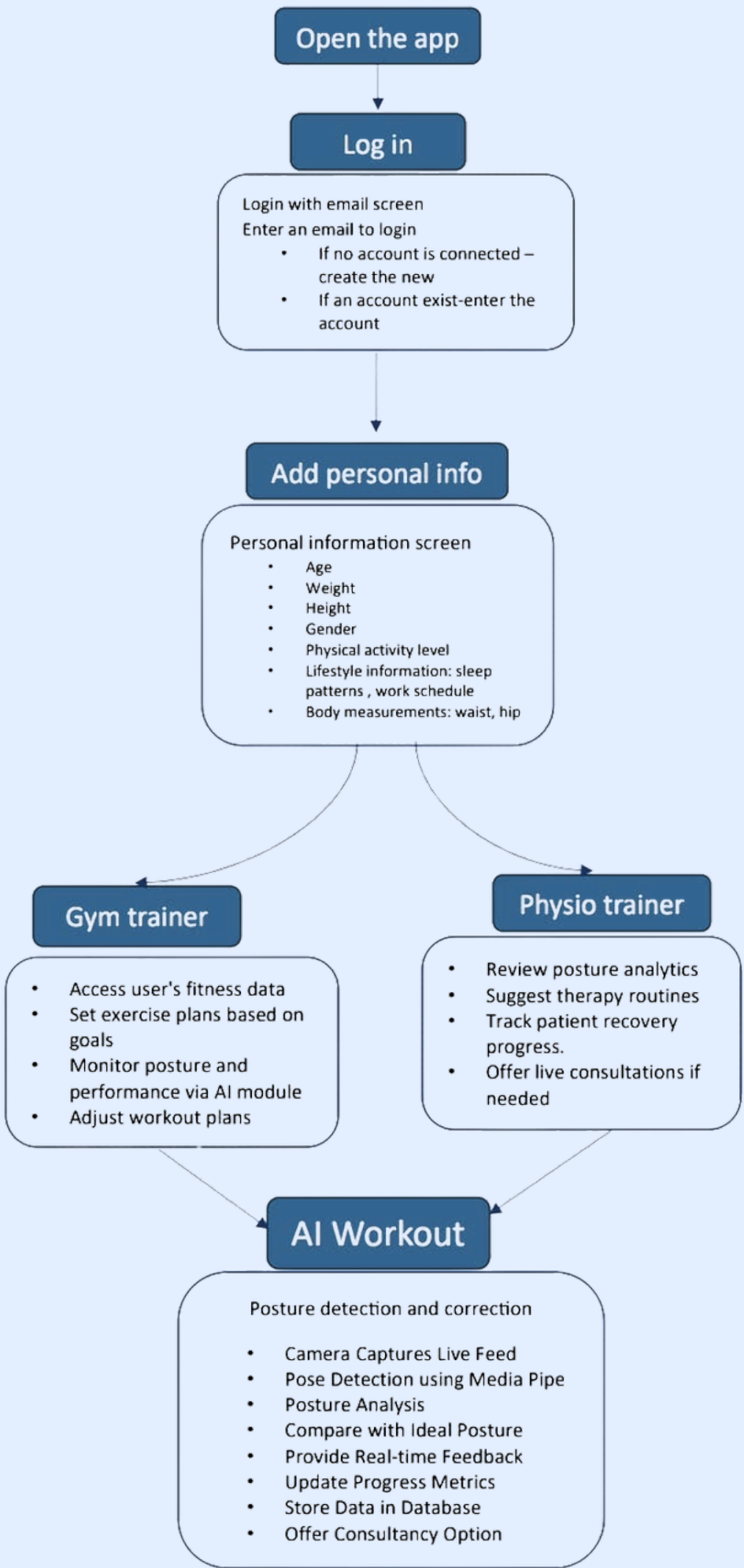
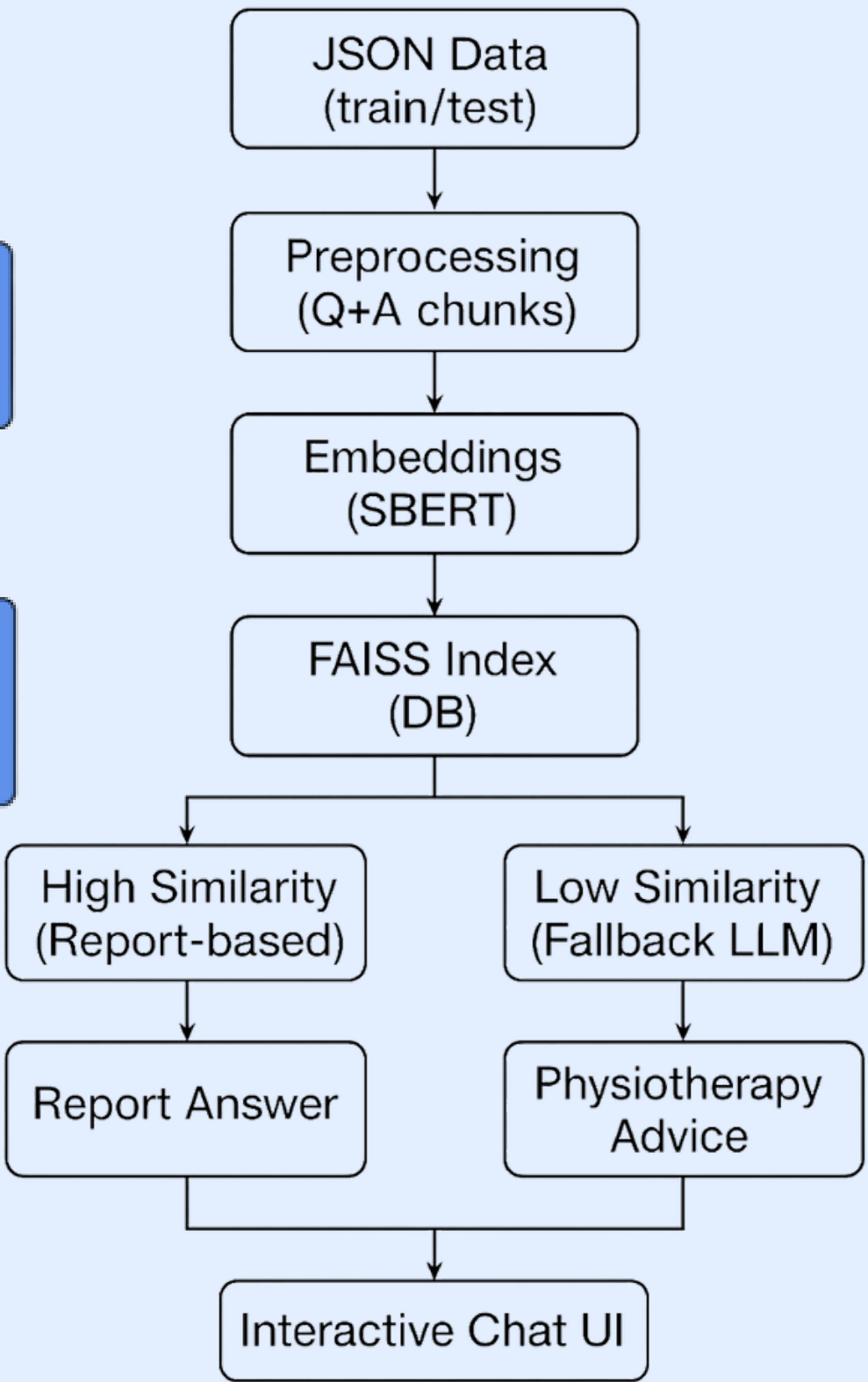
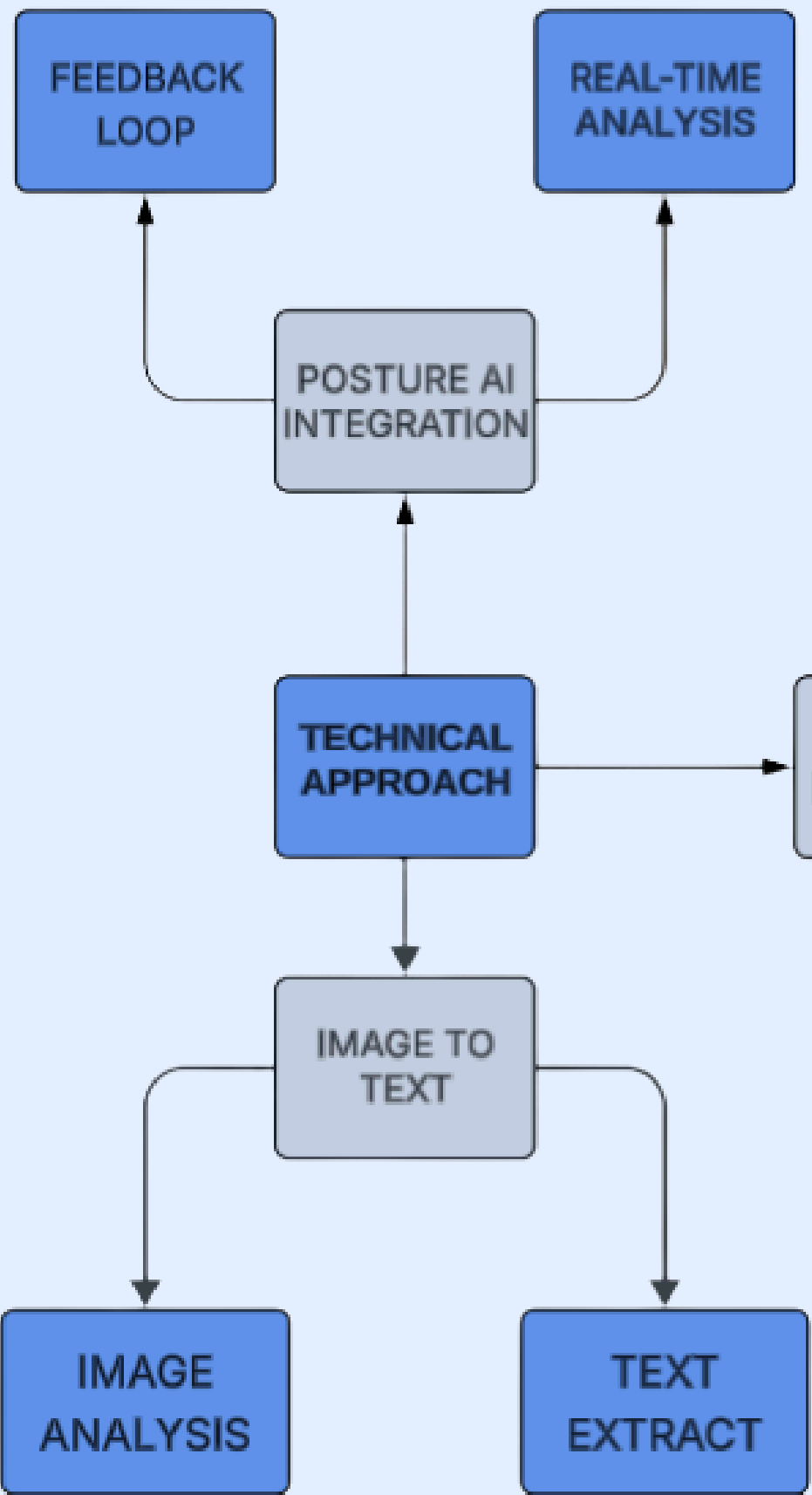
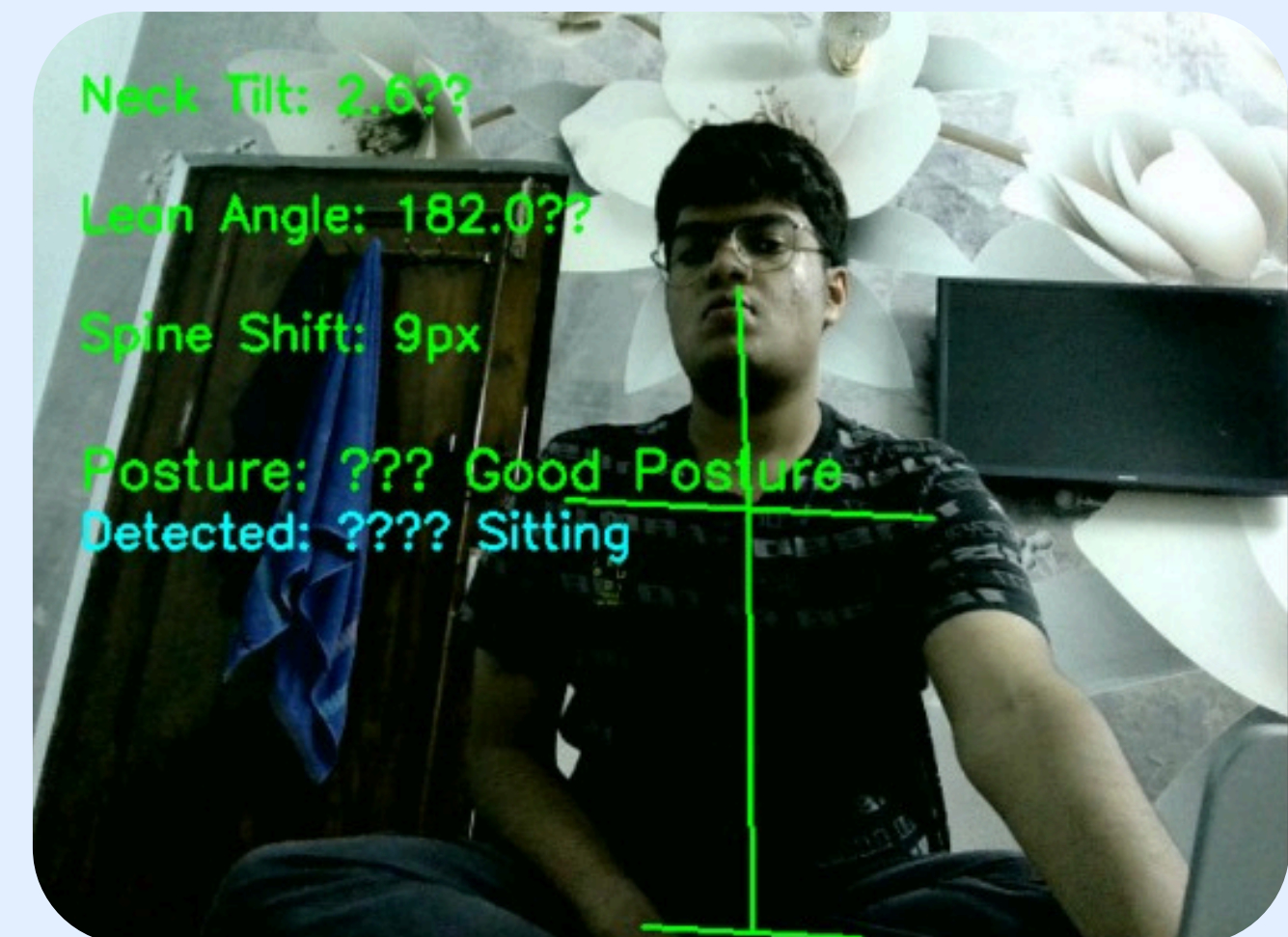
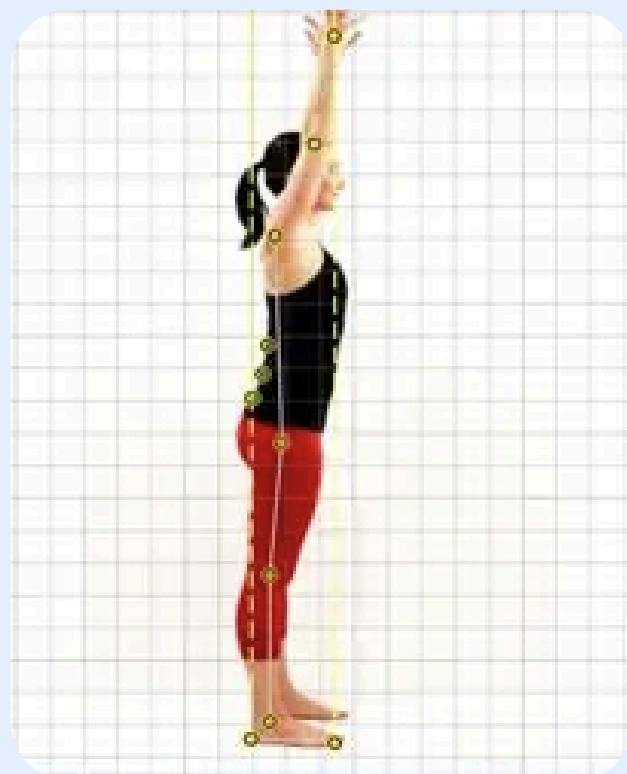
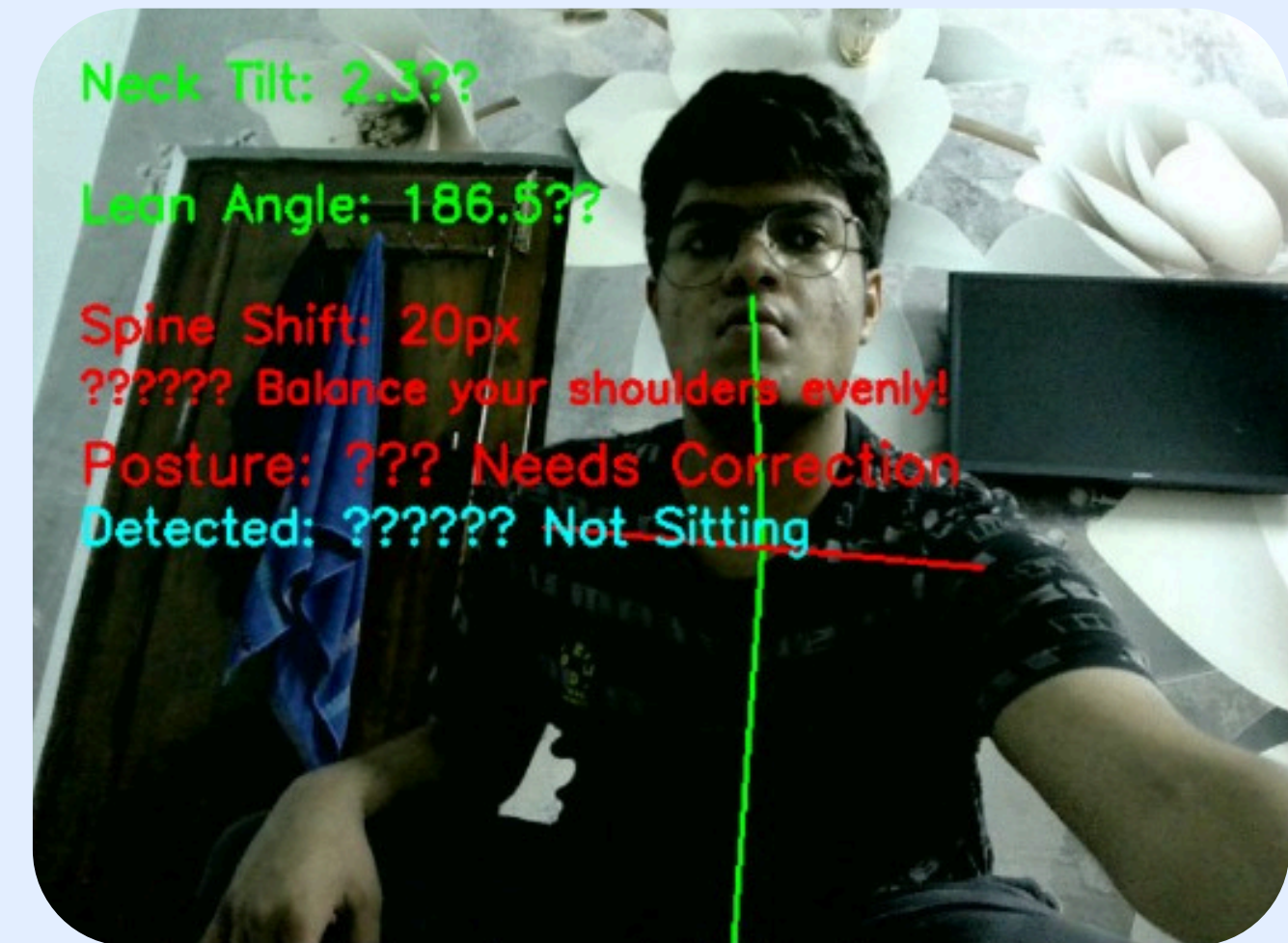
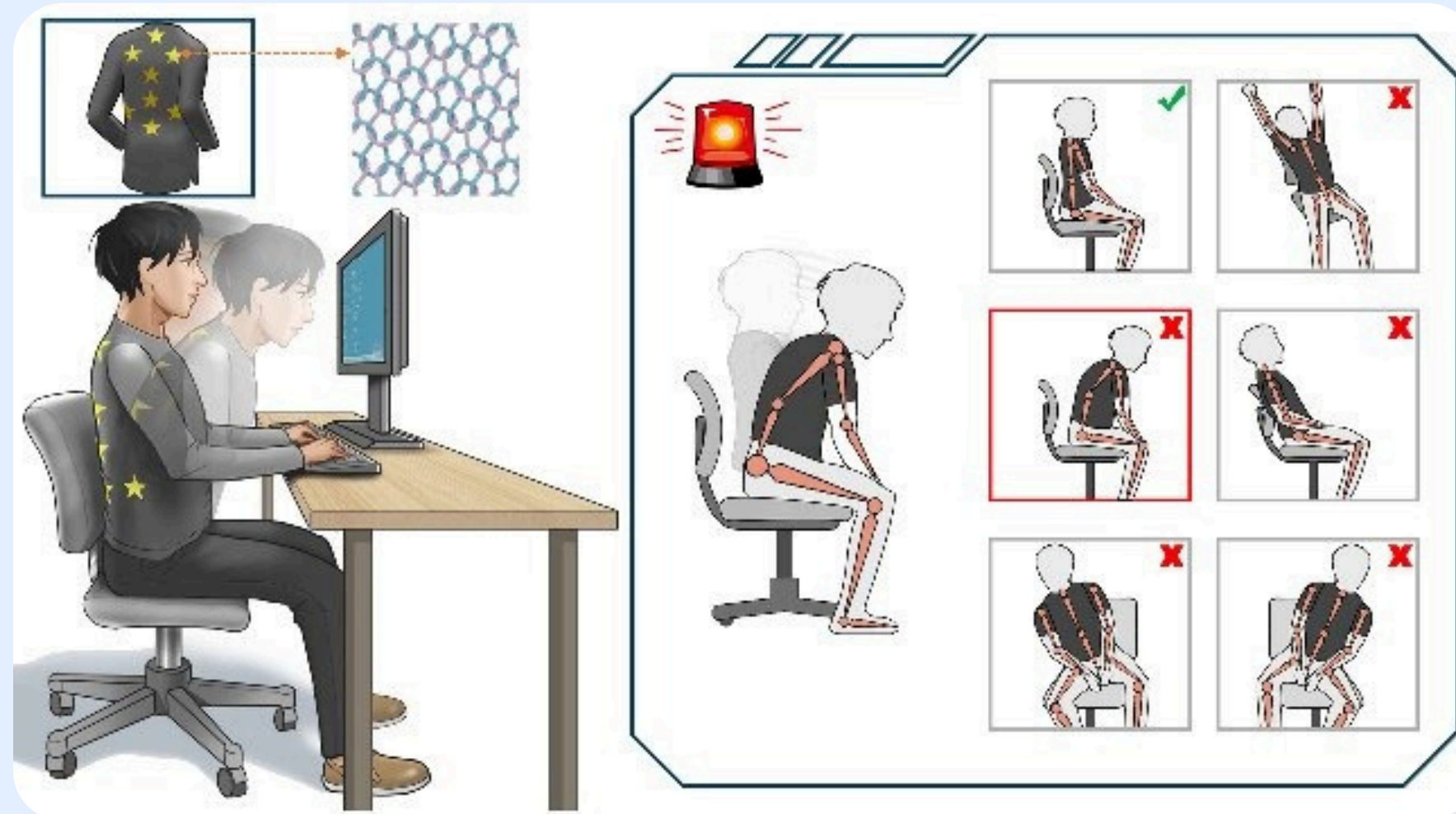


IMAGE ANALYSIS



Evolving Trends of Fitness & Physiotherapy

Global Fitness



Indian Fitness



Hybrid Workout



01

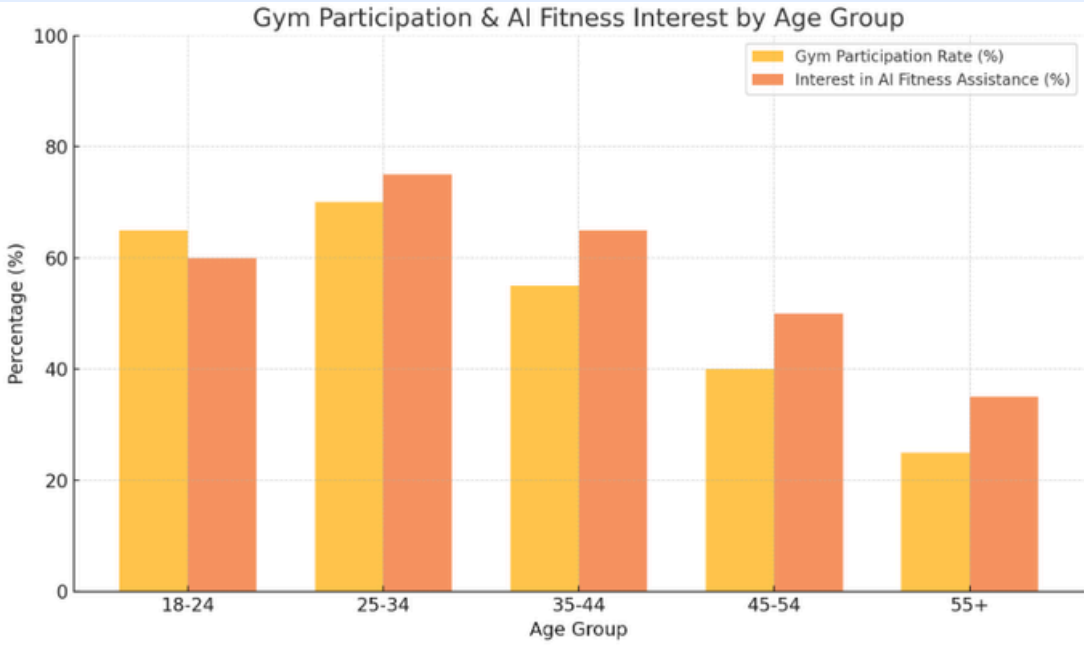
The global fitness and gym market is expected to reach \$135 billion by 2028 (Harvard university)

02

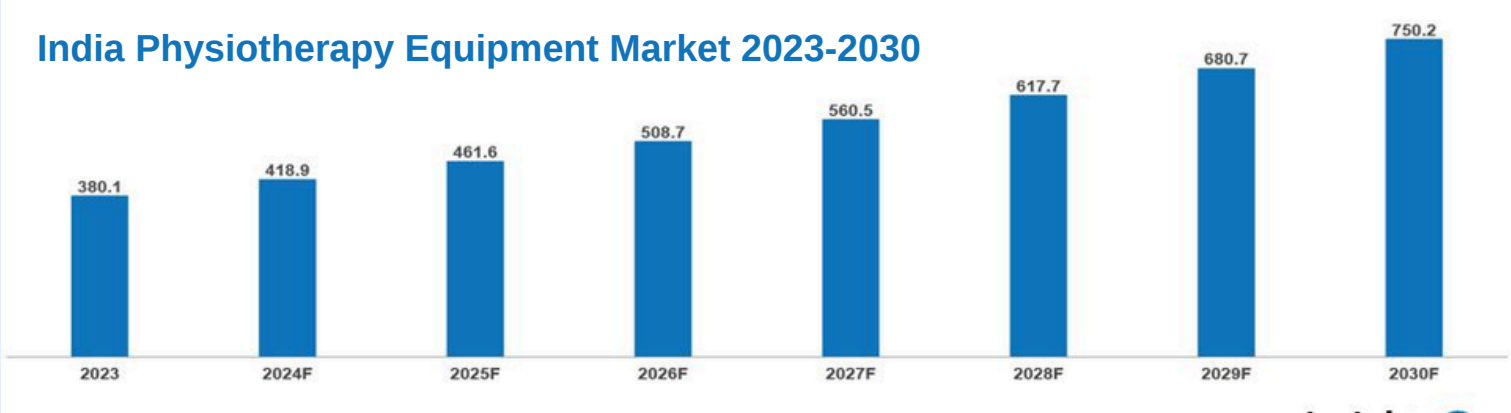
India has over 6 million gym-goers (ages 18–45), a key segment for adoption (WHO data).

03

Home and hybrid workout trend: ~58% of gym members also engage in at-home workouts (Statista, 2024).



Age Group	Gym Participation	AI Fitness Interest (%)
18-24	65%	60%
25-34	70%	75%
35-44	55%	65%
45-54	40%	50%
55+	25%	35%



THANKYOU

Experience the future of healthcare with our AI-powered platform that combines physiotherapy monitoring, medical report analysis, and telemedicine consultation in one accessible solution.

